How the Orioles Lost Too Many 'Should-Have-Won' Games in 2025

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Abstract

As fans, we remember the 2025 Orioles season as a disappointment, but why? How did a talented roster fall short of the playoffs? There are likely many contributing factors, but this analysis focuses on one specific angle: how often the Orioles lost games they were expected to win. By identifying these "should-have-won" games, we can estimate how much those losses impacted their season, and what needs to improve going into 2026.

Research Question

How do we define a "should-have-won" game?

I began with a rule-based approach, defining these games as ones in which the Orioles:

- Scored 5+ runs
- Held a 3+ run lead
- Led after the 8th inning

However, I expanded the analysis to include a more objective metric: **win probability**. Using data from MLB's Stats API, any game in which the Orioles reached at least an 80% win probability at any point but still lost was also classified as a "should-have-won" game.

Pipeline

Using the MLB Stats API, I parsed all **2,430 regular season games** from 2025.

Key steps:

- Extracted play-by-play win probability data for each game
- Removed terminal values (0% and 100%) to avoid skewing results
- Created opportunity metrics:
  - Games reaching X% win probability (80%, 85%, 90%, 95%)
  - Games leading after 8 innings
  - Games scoring 5+ runs
- Defined failures as losses after reaching those states
- Calculated **failure rates = failures / opportunities**

This allowed for a consistent comparison of how teams performed when put in strong winning positions.

League Context

To properly evaluate the Orioles, their performance must be compared to the rest of MLB.

Across these metrics, the Orioles were consistently below league average, though not always extreme outliers. However, teams that ranked poorly in these categories were rarely playoff teams.

The Orioles ranked in the top 10 worst teams in:

- 95% win probability loss rate
- 80% win probability loss rate
- Wasted offense (losses when scoring 5+ runs)

Figure 1 shows the distribution of 95% and 80% win probability failure rates across MLB, with the Orioles clearly positioned among the worst teams.

Notably, almost no playoff teams appeared in these bottom-10 groups, reinforcing that poor performance in these areas is strongly associated with missing the postseason.

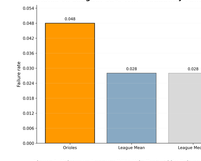

Figure 1. Orioles vs League: 95% Win Probability Failure Rate

Results

The Orioles lost too many games from winning positions.

- **15 losses** after reaching 80% win probability (7th most)
- **3 losses** after reaching 95% win probability (4th most)
- **22 rule-based losses**, slightly above the league average of 20.5

Figure 2 highlights how the Orioles compare to the league average in total "should-have-won" losses.

These are not just bad outcomes—they are high-leverage failures that swing seasons.

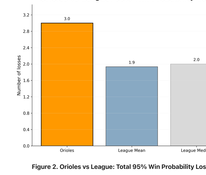

Figure 2. Orioles vs League: Total 95% Win Probability Losses

Opportunities

The Orioles also had fewer chances than average to be in strong positions:

- 23rd in games reaching 95% win probability
- 21st in games reaching 80%
- 24th in games scoring 5+ runs

Figure 3 shows that the Orioles ranked in the bottom third of MLB in high win probability opportunities.

This is important: the Orioles were not only worse at converting wins, but also worse at creating those opportunities in the first place.

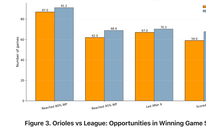

Figure 3. Orioles vs League: Opportunities in Winning Game States

Analysis

The data aligns closely with the eye test.

The Orioles:

- Did not consistently put themselves in position to win
- Failed to close out games when they did

This is likely tied to:

- Inconsistent starting pitching
- Bullpen instability
- Injuries (only 3 players avoided injury all season)

But the key takeaway is measurable:

- **95% WP games:** Orioles win rate = **95.2%**, league average = **97.2% (5th worst)**
- **3+ run leads:** Orioles = **87.5%**, league = **90.6% (7th worst)**
- **80% WP games:** Orioles = **82.8%**, league = **86.1% (7th worst)**
- **5+ run games:** Orioles = **74.6%**, league = **79.5% (7th worst)**

Across nearly every metric, the Orioles consistently ranked in the bottom quartile of MLB.

Figure 4 visualizes the relationship between opportunities and failure rates, showing that the Orioles struggled both to create and convert winning positions.

The only area close to average was the 85% win probability category.

Figure 4. Opportunities vs Failure Rate Across MLB

Comebacks

I also analyzed the opposite scenario: comeback wins.

The Orioles ranked:

- Bottom 10 in nearly every comeback metric
- **29th (second worst)** in games where they fell to 20% win probability
- Orioles comeback rate at 20% WP: **7.4%**
- League average: **12.8%**

This indicates a lack of resilience when trailing.

Figure 5 shows the distribution of 20% comeback rates across MLB, where the Orioles rank 29th.

While the Orioles had 3 wins from 5% win probability—memorable moments—the overall impact was minimal, roughly a **1-game swing** over the season. In contrast, blown games represent a much larger gap.

A full breakdown of all metrics and rankings is shown in **Figure 6**, which summarizes the Orioles' performance across collapse, opportunity, and comeback categories.

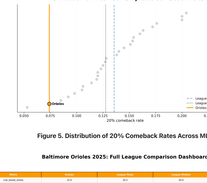

Figure 5. Distribution of 20% Comeback Rates Across MLB

Baltimore Orioles 2025: Full League Comparison Dashboard

Metric	Orioles	League Mean	League Median	Rank
rule_based_losses	22.0	20.5	20.0	9
loss_95_after_win	3.0	1.9	2.0	4
loss_95_losses	3.0	1.9	2.0	4
loss_95_losses	1.0	0.5	1.0	7
loss_80_losses	15.0	7.8	7.0	7
games_reached_95_pct	95.2	97.2	100.0	24
games_reached_95_wp	61.4	66.1	64.0	23
games_reached_80_pct	81.0	91.2	97.5	21
games_reached_80_wp	91.2	86.1	87.0	7
game_reached_led_8_pct	70.5	70.1	71.5	16
games_scored_5_plus	59.5	67.7	69.5	24
rule_based_win_pct	0.333	0.333	0.333	10
loss_95_pct_loss	0.115	0.103	0.100	6
loss_95_pct_loss	0.115	0.103	0.100	6
loss_80_pct_loss	0.600	0.506	0.500	10
loss_80_pct_loss	0.556	0.524	0.529	7
loss_rate_loss_rate	0.136	0.127	0.123	10
big_lead_loss_rate	0.133	0.098	0.098	7
comeback_win_rate_20	0.074	0.128	0.125	29
loss_95_rate	0.152	0.108	0.100	7
loss_95_loss_rate	0.150	0.096	0.095	11
loss_80_rate	0.682	0.506	0.500	10
loss_80_loss_rate	0.682	0.510	0.519	7
games_lead_8_pct	70.5	70.1	71.5	16
games_lead_8_95_wp	95.2	91.8	95.7	19
games_lead_8_80_wp	91.2	86.0	91.3	8
rule_based_losses_pct	0.667	0.621	0.606	9
loss_95_comeback_pct	0.5	0.5	0.5	18
loss_95_comeback_pct	0.5	0.5	0.5	18
loss_95_comeback_pct	1.0	1.0	1.0	1
loss_95_comeback_pct	0.667	0.667	0.667	1
loss_95_comeback_rate	0.667	0.646	0.646	26
loss_95_comeback_rate	0.646	0.657	0.666	26
loss_95_comeback_rate	0.556	0.657	0.629	9
games_lead_wp_after_8	70.5	70.1	69.5	8
losses_after_8_pct	0.6	0.7	0.7	8
losses_after_8_pct_rate	0.136	0.125	0.127	10

Figure 6. Orioles 2025 Full Metric Summary vs League

Conclusion

The Orioles' 2025 season was not defined by a lack of talent; it was defined by a failure to convert game states into wins.

They struggled in two critical areas:

1. **Protecting leads** — ranking among the worst teams in converting strong win probabilities
2. **Recovering from deficits** — ranking near the bottom in comeback success

Even a modest improvement in these areas could have significantly changed the outcome of the season. Converting just a handful of high-probability losses into wins likely would have been enough to push the Orioles into playoff contention.

For 2026, the path forward is clear:

- Improve *pitching stability* to protect leads
- Increase *offensive consistency* to create more opportunities
- Develop late-game execution, both ahead and behind

Because in a 162-game season, the difference between missing the playoffs and making a run often comes down to a small number of games—and for the Orioles, those were the games they should have already won.

All charts and full data outputs are included in the Appendix and supplementary materials.

Appendix

A1. Data Sources

All data was obtained using the MLB Stats API, including:

- Game schedules and results
- Play-by-play data
- Win probability data for each event

A2. Definitions

Win Probability (WP)
The probability that a team will win the game at a given moment, based on game state.

Should-Have-Won Game
A loss in which a team reached a specified win probability threshold (80%, 85%, 90%, or 95%) or met rule-based criteria (5+ runs, 3+ run lead, or leading after 8 innings).

Opportunity Metrics
The number of games in which a team reached a specific game state (for example, 90% WP or leading after 6 innings).

Failure Rate
Failures divided by opportunities:

```
failure rate = losses / opportunities
```

Comeback Rate
Wins divided by opportunities when trailing:

```
comeback rate = comeback wins / games in trailing state
```

A3. Methodology Notes

- Terminal win probabilities (0% and 100%) were excluded to avoid bias
- All metrics were calculated at the team level
- League averages were computed as the **mean of team-level rates**, ensuring equal weighting across teams
- Only regular season games (162 per team) were included

A4. Limitations

- Win probability does not capture all contextual factors (for example, bullpen strength or matchups)
- Some late-game states may be disproportionately impactful
- Results are descriptive and do not establish causation

Supplementary Materials

All data outputs, processed datasets, and visualizations generated for this analysis are included as supplementary materials. Due to platform limitations, the full `charts` directory (containing all figures) is not embedded directly within this notebook.

The supplementary materials include:

- The full team-level dataset (`mlb_2025_should_have_won_team_summary.csv`)
- Game-level logs for all 2,430 MLB games analyzed
- Orioles vs. league comparison tables
- Over 40 generated visualizations covering collapse metrics, opportunity metrics, and comeback performance

The `charts` directory contains all figures referenced in this article, including:

- Win probability failure rate comparisons
- Opportunity vs. conversion scatter plots
- League-wide rankings and distributions
- Comeback win and rate analyses
- A full Orioles performance dashboard

All materials are reproducible using the provided Python scripts. For full access to the visual outputs, please download the repository and accompanying files where the `charts` folder is included in its entirety.

Supplementary materials (all scripts and figures): Download orioles-2025-win-prob-analysis.zip